

Mathematics I

Week 3

Graded Assignment Practice Session

Plan for this session

- How to join?
 - Join on webex - click on link sent to you
 - Join on pear deck - joinpd.com (enter code: see top right on webex)
 - Keep a notebook and pen ready for solving problems
- For every question - 5 to 15 minutes allotted
 - Question will be shown in a slide for solving
 - If you are done solving, enter your answer on joinpd.com
 - Presenter will provide a solution
 - Questions and discussion

Example Screenshots

Laptop/Desktop

The screenshot shows a presentation slide with a dark background. On the left, there is a large grey rectangular area labeled "Q1 (a)". To its right, the text reads: "Is this function even or odd or neither even nor odd?". In the top right corner, a small box contains the text "How to participate? joinpd.com code". In the bottom left corner, there is a small logo with the letters A, B, and C, and the text "Students choose an option". In the bottom right corner, it says "Pear Deck Interactive Slide Do not remove this bar".

Portion for Answering

☐ Even

☐ Odd

☐ Neither even nor odd

Mobile

The screenshot shows the same presentation slide on a mobile phone screen. The top status bar shows the time as 2:50 PM and various icons. The address bar shows "app.peardeck.com/studi". The slide content is the same as the laptop/desktop version. A red circle highlights the "Answer Question" button at the bottom right of the slide.

Portion for Answering

☐ Even

☐ Odd

☐ Neither even nor odd

Answer Question

Test Problem

How to participate?
joinpd.com
code: see above

This is a problem to test if everything is working.

Have you attended the Paradox 2022?



Students, write your response!

“ It is not being BEST!

It is being better than yesterday”

Q1

Suppose $ax^2 + bx + c = 0$ is a quadratic equation. If the sum of the roots is +4 and the product of the roots is -12.

Q1

Suppose $ax^2 + bx + c = 0$ is a quadratic equation. If the sum of the roots is +4 and the product of the roots is -12.

Find the value of $\frac{4b+3c}{a}$.



Students, write your response!

Solution

Step-1:

The sum of the roots of $ax^2 + bx + c = 0$ is $-b/a$.

$$\therefore \frac{-b}{a} = 4 \Rightarrow \boxed{\frac{b}{a} = -4}$$

The Product of the roots is c/a .

$$\therefore \boxed{\frac{c}{a} = -12}$$

Solution

Step-2:

The value of $\frac{4b+3c}{a} = 4\left(\frac{b}{a}\right) + 3\left(\frac{c}{a}\right) = 4(-4) + 3(-12) = -16 - 36 = -52$.

Solution

Step-2:

The value of $\frac{4b+3c}{a} = 4\left(\frac{b}{a}\right) + 3\left(\frac{c}{a}\right) = 4(-4) + 3(-12) = -16 - 36 = -52$.

Answer: -52

Q2(a)

The slope of a parabola $y = 3x^2 - 8x + 5$ at a point (p, q) is 4.

Q2(a)

The slope of a parabola $y = 3x^2 - 8x + 5$ at a point (p,q) is 4.

Find the value of p .



Students, enter a number!

Solution

Step 1:

The slope of a parabola $y = ax^2 + bx + c$ at a point (x_0, y_0) is $2ax_0 + b$.
Here, $a = 3$, $b = -8$, $c = 5$, $x_0 = p$, $y_0 = q$.

Solution

Step 1:

The slope of a parabola $y = ax^2 + bx + c$ at a point (x_0, y_0) is $2ax_0 + b$.
Here, $a = 3$, $b = -8$, $c = 5$, $x_0 = p$, $y_0 = q$.

Step 2:

Substituting these values, we get

The slope of a parabola $y = 3x^2 - 8x + 5$ at a point (p, q) is $2(3)(p) - 8 = 6p - 8$.

So, $6p - 8 = 4 \implies \mathbf{p = 2}$.

Answer: **2**

Q2(b)

The slope of a parabola $y = 3x^2 - 8x + 5$ at a point P is 4.

Q2(b)

The slope of a parabola $y = 3x^2 - 8x + 5$ at a point P is 4.

Find the sum of the x-coordinate and y-coordinate of the point P.



Students, enter a number!

Solution

Step 1:

Let the point P be (p,q) . Here, p is the x - coordinate and q is the y -coordinate of the point P.

We have to find $p + q$.

Solution

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Let the point P be (p,q) . Here, p is the x- coordinate and q is the y-coordinate of the point P.

We have to find $p + q$.

Step-2:

From the previous question, we know that $p = 2$.

Solution

Step 3:

As (p, q) is a point on the parabola $y = 3x^2 - 8x + 5$, it should satisfy the given equation.
Hence, $q = 3p^2 - 8p + 5 = 3(2)^2 - 8(2) + 5 = 1$.

Solution

Step 3:

As (p, q) is a point on the parabola $y = 3x^2 - 8x + 5$, it should satisfy the given equation.
Hence, $q = 3p^2 - 8p + 5 = 3(2)^2 - 8(2) + 5 = 1$.

Step-4:

Therefore, the value of $p + q = 2 + 1 = 3$.

Answer: 3

Q3

A ball is thrown off from the top of a building. The height of the ball(in feet) from the ground after 't' seconds is given by

$$H(t) = -2t^2 + 4t + 16.$$

Q3

A ball is thrown off from the top of a building. The height of the ball(in feet) from the ground after 't' seconds is given by

$$H(t) = -2t^2 + 4t + 16.$$

NOTE: Consider the foot of the building at the origin.

Which of the following statements are correct?

- (a) The ball reaches its maximum height after 1 second.
- (b) The height of the building is 16 feet.
- (c) The ball travels in the air for 4 seconds before it reaches the ground.
- (d) The maximum height reached by the ball is 18 feet.



Students, write your response!

Solution

Step 1:

Draw a rough sketch of the data given.

Solution

Step 1:

Draw a rough sketch of the data given.

Step-2:

To draw the rough graph of the parabola $-2t^2 + 4t + 16$, we need to

- Find the vertex of the parabola.
- Find the y-intercept of the parabola.
- Find the roots of the parabola(x-intercepts).

Solution

Step 3:

- The vertex of the given parabola will be at (1,18).

$$x = \frac{-b}{2a} = \frac{-(4)}{2(-2)} = 1 \quad \& \quad y = c - \frac{b^2}{4a} = 16 - \frac{4^2}{4(-2)} \\ = 18$$

Solution

Step 3:

- The vertex of the given parabola will be at (1,18).

$$x = \frac{-b}{2a} = \frac{-(4)}{2(-2)} = 1 \quad \& \quad y = c - \frac{b^2}{4a} = 16 - \frac{4^2}{4(-2)} \\ = 18 .$$

- The y-intercept will be at (0,16).

Substituting $t = 0$, we get

$$H(0) = -2(0)^2 + 4(0) + 16 = 16 .$$

Solution

Step 3:

- The vertex of the given parabola will be at (1,18).

$$x = \frac{-b}{2a} = \frac{-(4)}{2(-2)} = 1 \quad \& \quad y = c - \frac{b^2}{4a} = 16 - \frac{4^2}{4(-2)} = 18.$$

- The y-intercept will be at (0,16).

Substituting $t = 0$, we get

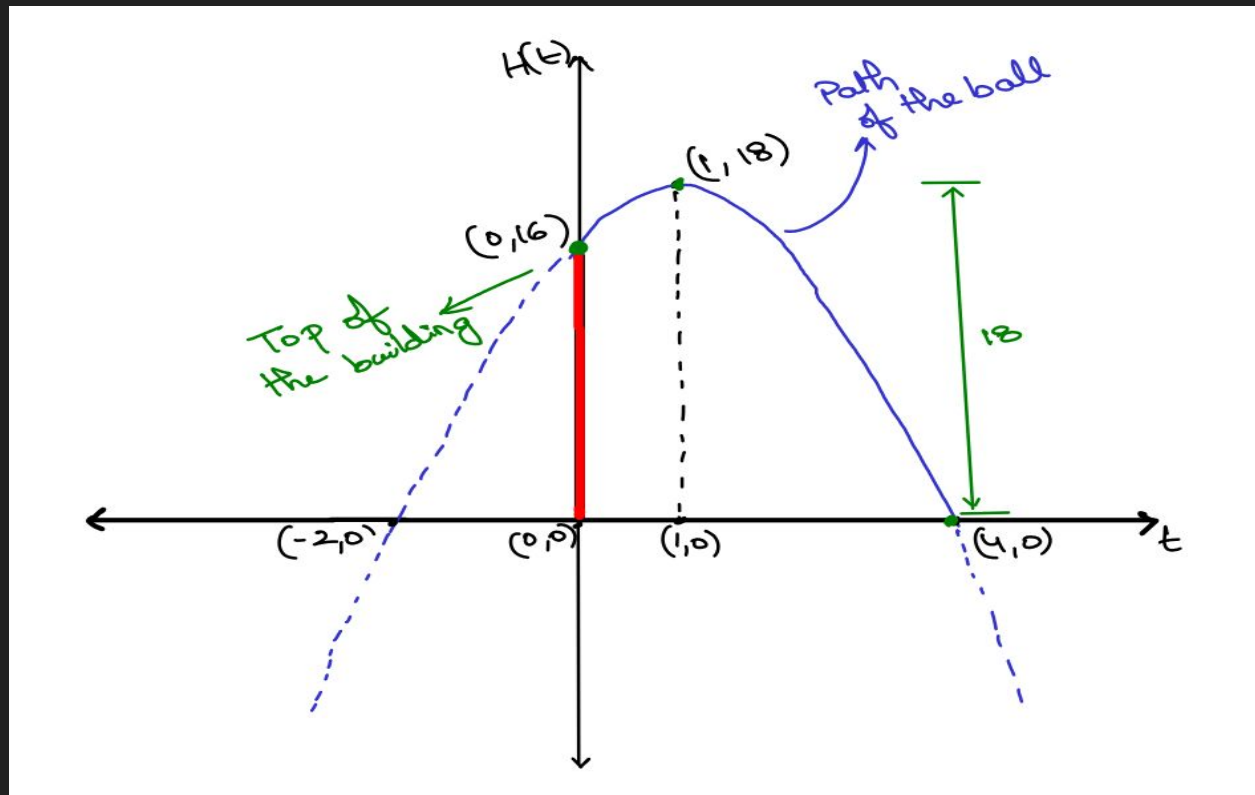
$$H(0) = -2(0)^2 + 4(0) + 16 = 16.$$

- The roots of the parabola are -2, 4.

$$\begin{aligned} -2t^2 + 4t + 16 = 0 &\Rightarrow t^2 - 2t - 8 = 0 \Rightarrow t^2 - 4t + 2t - 8 = 0 \\ &\Rightarrow t(t-4) + 2(t-4) = 0 \Rightarrow (t+2)(t-4) = 0 \Rightarrow t = -2 \text{ \& } t = 4 \end{aligned}$$

Solution

Step 4: The rough graph is shown below:



Solution

All the options are correct.

Answer: (a),(b),(c),(d)

Q4

Suppose $y = \frac{x^2}{2}$ is the equation of a parabola and $4y - x = 3$ is a straight line.

Q4

Suppose $y = \frac{x^2}{2}$ is the equation of a parabola and $4y - x = 3$ is a straight line.

Find the points of intersection of the given parabola and the straight line.

- (a) $(-3/2, 9/8)$
- (b) $(1, 1/2)$
- (c) $(3/2, 9/8)$
- (d) $(-1, 1/2)$



Students, write your response!

Solution:

Step-1:

- Write the equation of the straight line in slope-intercept form.

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- $4y - x = 3 \implies y = \frac{x}{4} + \frac{3}{4}$

Solution:

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- Write the equation of the straight line in slope-intercept form.
- $4y - x = 3 \implies y = \frac{x}{4} + \frac{3}{4}$

Step-2:

- Equate both the equations, $y = \frac{x^2}{2}$ and $y = \frac{x}{4} + \frac{3}{4}$.

Solution:

Step-1:

- Write the equation of the straight line in slope-intercept form.
- $4y - x = 3 \implies y = \frac{x}{4} + \frac{3}{4}$

Step-2:

- Equate both the equations, $y = \frac{x^2}{2}$ and $y = \frac{x}{4} + \frac{3}{4}$.
- $\frac{x^2}{2} = \frac{x}{4} + \frac{3}{4} \implies 2x^2 - x - 3 = 0$

Solution:

Step-1:

- Write the equation of the straight line in slope-intercept form.
- $4y - x = 3 \implies y = \frac{x}{4} + \frac{3}{4}$

Step-2:

- Equate both the equations, $y = \frac{x^2}{2}$ and $y = \frac{x}{4} + \frac{3}{4}$.
- $\frac{x^2}{2} = \frac{x}{4} + \frac{3}{4} \implies 2x^2 - x - 3 = 0$

Step-3:

- Find the roots of the obtained quadratic equation to get the x-coordinates of the intersection points.

Solution:

Step-3 (cont.): Use factorization method to solve the quadratic equation.

Solution:

Step-3 (cont.): Use factorization method to solve the quadratic equation.

$$2x^2 - x - 3 = 0$$

$$2x^2 + 2x - 3x - 3 = 0$$

$$2x(x + 1) - 3(x + 1) = 0$$

$$(2x - 3)(x + 1) = 0$$

$$x = \frac{3}{2} \text{ or } -1$$

Solution:

Step-4:

- Substitute the values of 'x' in any of the equation (either quadratic or straight line) to get the y-coordinate of the intersection points.

Solution:

Step-4:

- Substitute the values of 'x' in any of the equation (either quadratic or straight line) to get the y-coordinate of the intersection points.

$$\text{If } x = \frac{3}{2},$$

$$y = \frac{x^2}{2} \implies y = \frac{(\frac{3}{2})^2}{2} \implies y = \frac{9}{8}$$

Therefore, the intersection point will be $(\frac{3}{2}, \frac{9}{8})$

Solution:

Step-4:

- Substitute the values of 'x' in any of the equation (either quadratic or straight line) to get the y-coordinate of the intersection points.

$$\text{If } x = \frac{3}{2},$$

$$y = \frac{x^2}{2} \implies y = \frac{(\frac{3}{2})^2}{2} \implies y = \frac{9}{8}$$

Therefore, the intersection point will be $(\frac{3}{2}, \frac{9}{8})$

$$\text{If } x = -1,$$

$$y = \frac{x^2}{2} \implies y = \frac{(-1)^2}{2} \implies y = \frac{1}{2}$$

Therefore, the intersection point will be $(-1, \frac{1}{2})$

Solution:

- Hence, there are two intersection points $(\frac{3}{2}, \frac{9}{8})$ and $(-1, \frac{1}{2})$

Answer: (c),(d)

Q5

The sum of squares of two consecutive even natural numbers is 340.

Q5

The sum of squares of two consecutive even natural numbers is 340. Find the product of those two even numbers.



Students, enter a number!

Solution:

Step-1:

- Let the two consecutive even natural numbers be ' $2k$ ' and ' $2k+2$ ', where ' k ' is any natural number.

Solution:

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- $(2k)^2 + (2k+2)^2 = 340$

Solution:

Step-1:

- Let the two consecutive even natural numbers be ' $2k$ ' and ' $2k+2$ ', where ' k ' is any natural number.
- $(2k)^2 + (2k+2)^2 = 340$

Step-2:

- We need to find the product of these two even natural numbers i.e., find the value of $(2k)(2k+2) = 4k^2 + 4k$.

Solution:

Step-3:

- Simplify the equation which is in step-1.

Solution:

Step-3:

- Simplify the equation which is in step-1.
- $(2k)^2 + (2k+2)^2 = 340 \Rightarrow 4k^2 + 4k^2 + 8k + 4 = 340$
 $\Rightarrow 8k^2 + 8k = 336$
 $\Rightarrow 4k^2 + 4k = 168$

Solution:

Step-3:

- Simplify the equation which is in step-1.
- $(2k)^2 + (2k+2)^2 = 340 \Rightarrow 4k^2 + 4k^2 + 8k + 4 = 340$
 $\Rightarrow 8k^2 + 8k = 336$
 $\Rightarrow 4k^2 + 4k = 168$

Answer: 168

Solution:

Step-3:

- Simplify the equation which is in step-1.
- $(2k)^2 + (2k+2)^2 = 340 \Rightarrow 4k^2 + 4k^2 + 8k + 4 = 340$
 $\Rightarrow 8k^2 + 8k = 336$
 $\Rightarrow 4k^2 + 4k = 168$

Answer: 168

Note: In this question, there is NO need to find the two natural numbers but in some cases you have to find them.

THANK YOU!!